

Week 8 : The Role of Governments – Externalities and Public Goods

(based on Chapter 9 of the text)

Externality

- (1) When a transaction between a buyer and a seller directly affects a third party, the effect is called an externality.
- (2) An externality is the cost or benefit that affects a party who did not choose to incur that cost or benefit.
- (3) An externality is an effect of a purchase or use decision by one set of parties on others who did not have a choice and whose interests were not taken into account.

A negative externality will arise when some of the costs of an activity (production or consumption) incur to those not directly involved.

An external cost is an uncompensated cost that an individual or firm imposes on others.

A positive externality will arise when some of the benefits of an activity are enjoyed by those not directly involved.

An external benefit is a benefit that an individual or firm gives to others without receiving compensation.

Adam Smith's "invisible hand" of the marketplace leads self-interested buyers and sellers in a market to maximize the total benefit that society can derive from a market.

But market failures can still happen.

EXTERNALITIES AND MARKET INEFFICIENCY

- Externalities cause markets to be inefficient, and thus fail to maximize total surplus.
- An externality arises...
 - . . . when a person engages in an activity that influences the well-being of a bystander and yet neither pays nor receives any compensation for that effect.

Negative Externality

In Production

Air pollution

Water Pollution

Greenhouse Gas emission

Noise pollution by Industry

Risks by Banking System

Overuse of antibiotics by farm animal production

In Consumption

Loud music late at night

Increased usage of antibiotics

Smoking or Alcohol abuse

Traffic congestion

Positive Externality

In Production

- Bee keepers

- First aid training by a firm

- Foreign firm demonstrates new technologies

- Research and Development (R&D) by a company

In Consumption

- An individual receiving a vaccination

- An individual maintaining attractive house

- Driving an electric cars

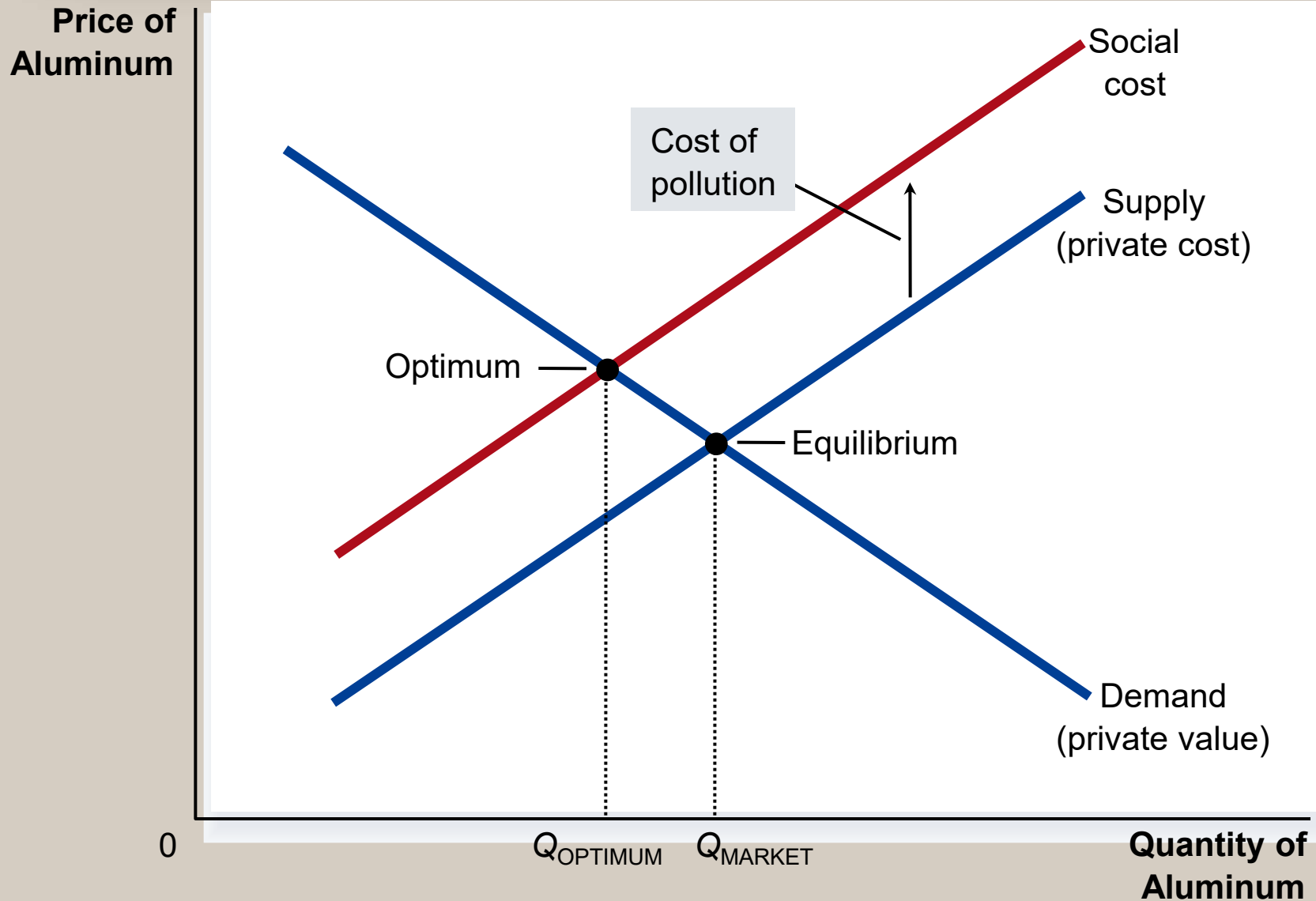
- Increased education of individuals

- Girl students in the class

EXTERNALITIES AND MARKET INEFFICIENCY

- Negative externalities lead markets to produce a larger quantity than is socially desirable.
- Positive externalities lead markets to produce a smaller quantity than is socially desirable.

Negative Externality and the Social Optimum



The Market for Aluminum

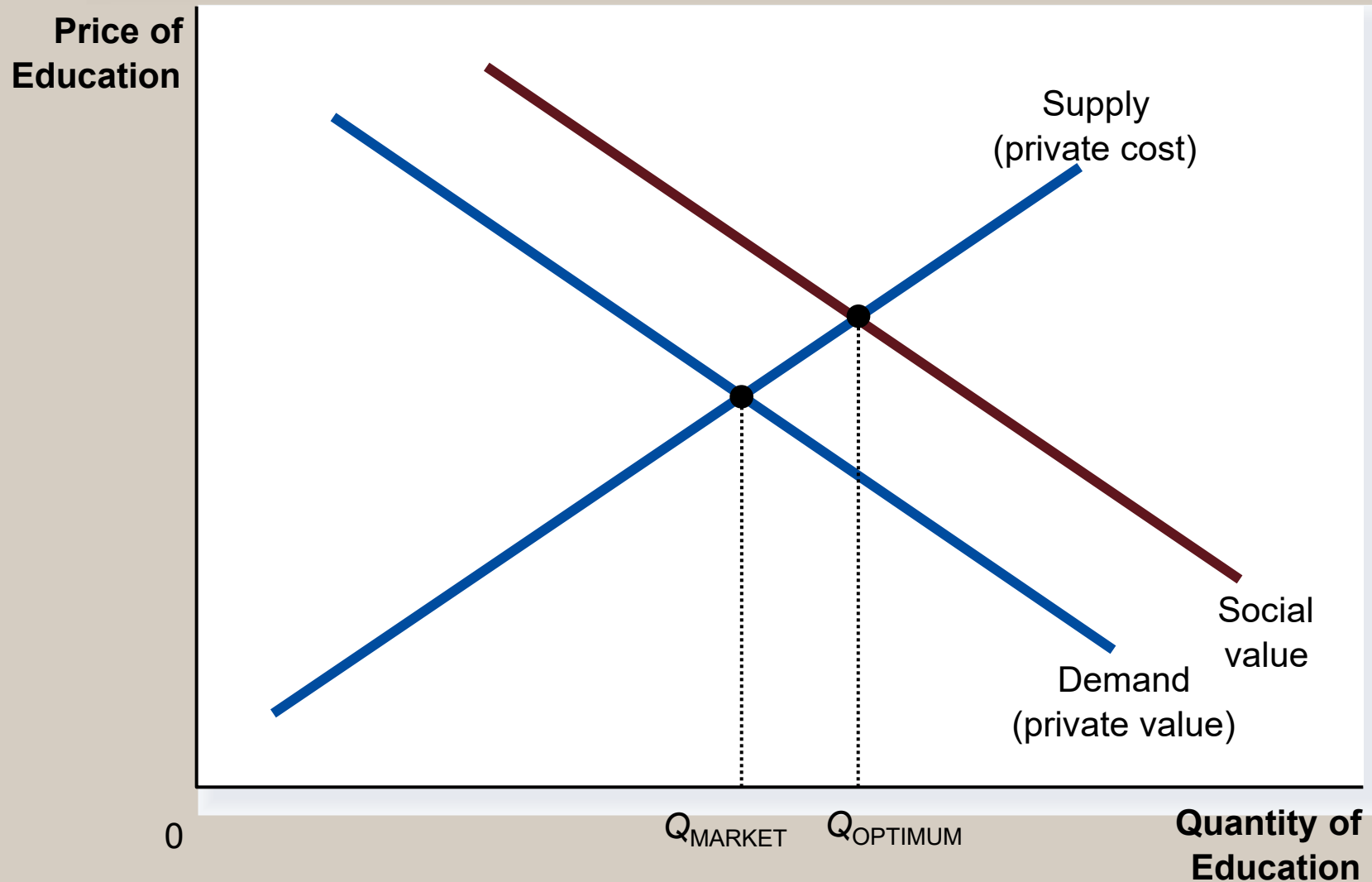
- For each unit of aluminum produced, the *social cost* includes the private costs of the producers plus the cost to those bystanders adversely affected by the pollution.
- The socially optimal output level *is less than* the market equilibrium quantity.

Achieving the Socially Optimal Output

Internalizing an externality involves altering incentives so that people take account of the external effects of their actions.

The government can internalize an externality by imposing a tax on the producer to reduce the equilibrium quantity to the socially desirable quantity.

Positive Externality and the Social Optimum



- The social value of the good exceeds the private value.
- The optimal output level is more than the market equilibrium quantity.
- The market produces a smaller quantity than is socially desirable.

Internalizing Externalities by Subsidies

- Used as the primary method for attempting to internalize positive externalities.

Industrial Policy

- Government intervention in the economy that aims to promote technology-enhancing industries
 - *Patent laws* are a form of technology policy that give the individual (or firm) with patent protection a *property right* over its invention.

PRIVATE SOLUTIONS TO EXTERNALITIES

Government action is not always needed to solve the problem of externalities.

The *Coase Theorem* is a proposition that if **transaction cost** is not so high, private parties can bargain and they can solve the problem of externalities on their own.

Transaction costs are the costs that parties incur in the process of agreeing to and following through on a bargain.

Example of Coase Theorem

- Karaoke shop makes noise
- Neighbors suffers from the noise

When the legal right is for the Karaoke shop

Neighbors may collect money and give the karaoke shop to low the noise if the benefit from the low noise $>$ money collected

When the legal right is for the neighbors

Karaoke shop may compensate neighbors for the noise when the benefit from making noise $>$ cost of the compensation

Karaoke Noise Solution

Karaoke has right

Karaoke benefit

Neighbors cost

400 Solution: Neighbors may
600 pay 500 to Karaoke

Karaoke benefit

Neighbors cost

800 Solution: Don't do anything
600 (Karaoke requires 800, but
neighbors do not want to pay
more than 600)

Neighbors has right

Karaoke benefit

Neighbors cost

600 Solution: Karaoke pays 500 to neighbors
400

Public Policy Toward Externalities

- The private solution approach can fail because transaction costs is too high for private agreement to be successful
- When externalities are significant and private solutions are not found, government may attempt to solve the problem through . . .
 - (1) command-and-control policies.
 - (2) market-based policies.

(1) Command-and-Control Policies

Usually take the form of regulations:

- Forbid certain behaviors.
- Require certain behaviors.

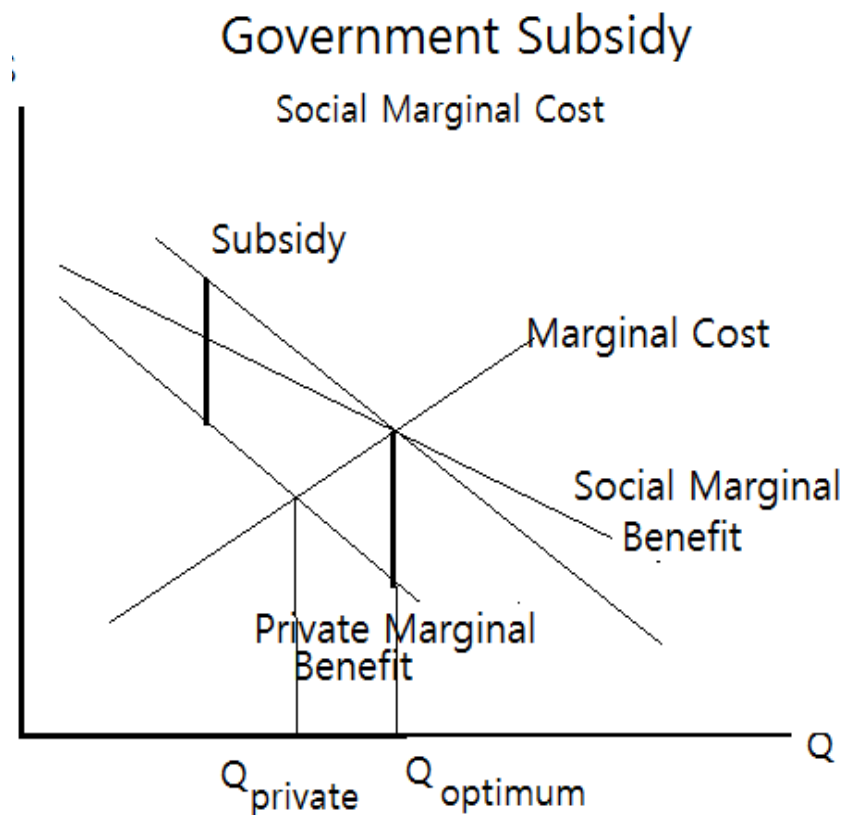
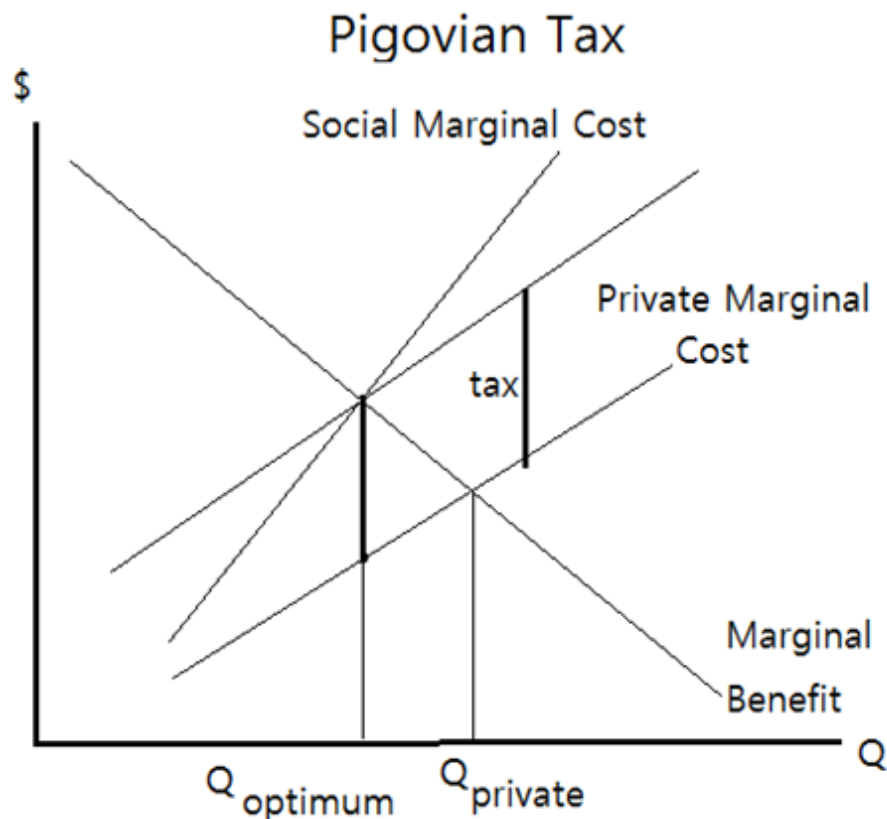
Examples:

- Requirements that all students be immunized.
- Limiting pollution emission levels by the Environmental Protection Agency (EPA).

(2) Market-Based Two Policies

- 1) Government uses taxes and subsidies to align private incentives with social efficiency.

Pigovian taxes are taxes enacted to correct the effects of a negative externality.



2) Tradable pollution permits allow trade of the right to pollute from one firm to another.

A firm that can reduce pollution at a low cost may prefer to sell its permit to a firm that can reduce pollution at a high cost.

Another Market Failure: The Case of Public Goods

Private Goods and Public Goods

A public good is both **nonexcludable** and **nonrival** in consumption.

ex) National Defense, Guiding Light

When a good is nonexcludable, the supplier cannot prevent consumption by people who do not pay for it.

Goods that are nonexcludable suffer from the free-rider problem: individuals have no incentive to pay for their own consumption and instead will take a “free ride” on anyone who does pay.

A good is nonrival in consumption

if more than one person can consume the same unit of the good at the same time. It does not reduce the remaining amount of goods or services

Characteristics of Private Goods

Excludable: No pay, No get

Rival: If I have it, no left for you.

Non-excludable : National defense, Guiding light, Fishing in the open ocean

Non-rival : TV programs, Guiding light

Market Failure: The Case of Public Goods

Providing Public Goods

If goods are non-excludable, a free rider problem occurs. This will cause inefficient low production

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If it is non-rival, the marginal cost becomes zero. Then the socially efficient price should be zero. No private firm will do business under this condition.

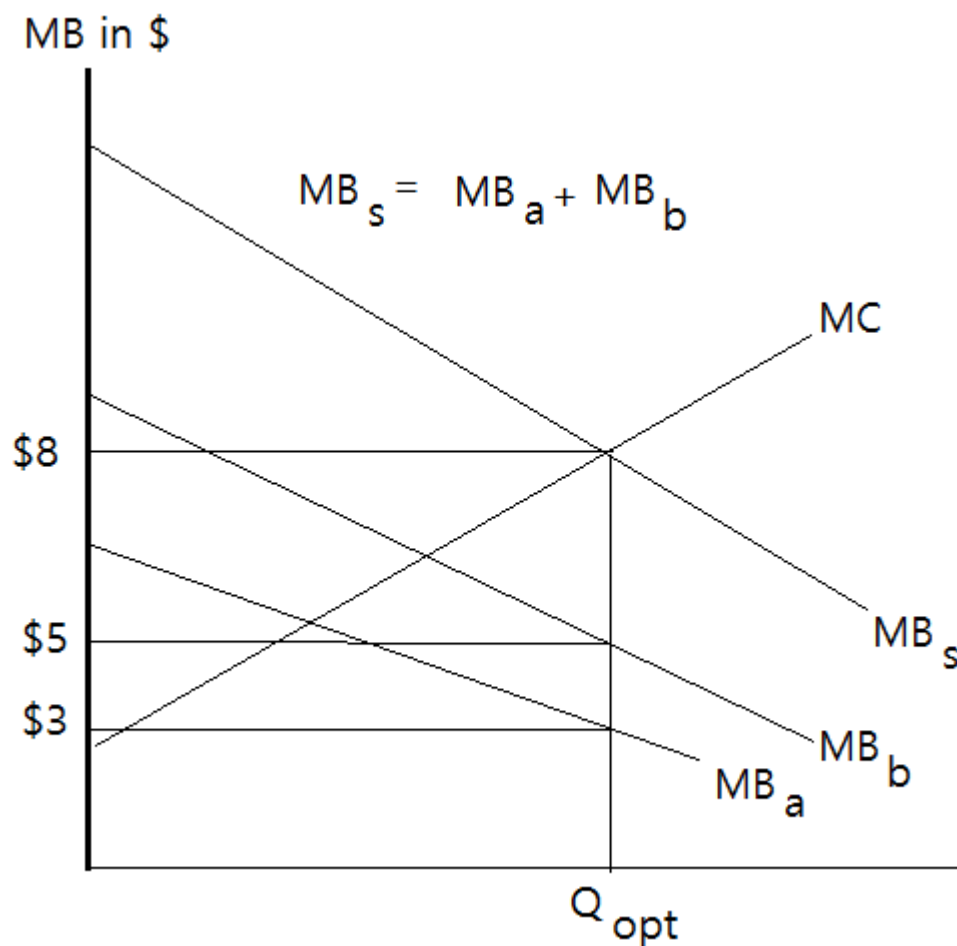
Some public goods may be supplied through voluntary contributions, or by private firms if they can earn money in an indirect way. Otherwise, government provides public goods, and this is one of the crucial roles of government.

Public Goods: How Much Should Be Provided?

The social marginal benefit is the sum of Mr. A's and Mr. B's marginal benefit. So the optimal quantity is Q_{opt} , and the price is \$8.

But Mr. A wants to pay only \$3, and Mr. B only \$5.

So we cannot rely on individual self-interest to come up with efficient quantity of public goods.



Summary

- When a transaction between a buyer and a seller directly affects a third party, the effect is called an externality.
- Negative externalities cause the socially optimal quantity in a market to be less than the equilibrium quantity.
- Positive externalities cause the socially optimal quantity in a market to be greater than the equilibrium quantity.

Summary

- Those affected by externalities can sometimes solve the problem privately.
- The Coase theorem states that if people can bargain without a cost, then they can always reach an agreement in which resources are allocated efficiently.

Summary

- When private parties cannot properly deal with externalities, then the government steps in.
- The government can either regulate behavior or internalize the externality by using Pigovian taxes or by issuing pollution permits.
- Private firms cannot make money from public goods. So government provides public goods.